

Dietary polyphenols in the prevention and treatment of allergic diseases.

Allergic disorders encompass skin, food and respiratory allergies. Sensitization to a normally harmless allergen results in the immune system being biased to a predominant T-helper type 2 response. Re-exposure to the same allergen leads to a robust secretion of allergy-related mediators that eventually triggers symptoms. Our understanding of these disorders has enabled the search of therapeutic approaches that can either modulate the sensitization process or impact on allergic mediators, thus helping manage allergic symptoms. Polyphenols are one such class of compounds that are found in foods and plant sources and have been investigated for their anti-allergic effect in different disease models and in human clinical trials. Their anti-inflammatory profile is known to impact on the recruitment of immune cells to the skin and in preventing the development of secondary infections following disruption of the skin barrier. The interaction of polyphenols with proteins can modulate the process of allergic sensitization and their direct effect on allergic effector cells such as mast cells inhibit mediator release, resulting in the alleviation of symptoms. In addition, their endogenous anti-oxidant ability limits the extent of cellular injury from free radicals during the allergic insult. Overall, polyphenols hold promise as anti-allergy agents capable of influencing multiple biological pathways and immune cell functions in the allergic immune response and deserve further investigation. The objective of the current review is to summarize the key findings and progress made in studying polyphenols as anti-allergic ingredients. Special emphasis is placed in this review to highlight key physiological, cellular and signalling pathways implicated in the mechanism of action of different polyphenols in the context of allergic disorders and their manifestations. © 2011 Blackwell Publishing Ltd.